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Oiowe chady

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Templatki (1)

.bashrc 11 lines

```
c() {
    g++ -std=c++20 -Wall -Wno-sign-conversion -Wextra -Wshadow -
        Wconversion -Wfloat-equal \
        -fsanitize=address,undefined -DLOCAL -DDEBUG -D_GLIBCXX_DEBUG
        -ggdb3 $1.cpp -o $1
}
```

```
nc() {
    g++ -std=c++20 -DLOCAL -O3 -static $1.cpp -o $1
}
```

```
alias cp='cp -i'
alias mv='mv -i'
```

.vimrc 7 lines

```
set nocompatible
filetype indent on
set autoindent
set number
syntax on
set ts=4 sw=4
set rnu
```

main.cpp 24 lines

```
#include <bits/stdc++.h>
using namespace std;
#ifdef DEBUG
auto&operator<<(auto &o, pair<auto, auto> p) {o << "(" << p.
    first << ", " << p.second << ")"; return o;}
auto operator<<(auto&o,auto x)->decltype(x.end(),o){o<<"{"; for
    (auto e : x) o<<e<<" "; return o<<"}";}
#define debug(X) cerr << "["#X"]": " << X << '\n';
#else
#define cerr if(0)cout
#define debug(X) ;
#endif
using ll = long long;
#define all(v) (v).begin(), (v).end()
#define ssize(x) int(x.size())
#define fi first
#define se second
#define mp make_pair
#define eb emplace_back
```

```
1 int main() {
    ios_base::sync_with_stdio(false); cin.tie(nullptr);

1     return 0;
}
```

gen.cpp
Opis: Dodatek do generowania testów, lub innego losowania. Ziarno można zmieniać np. `rng.seed(atoi(argv[1]))`;

```
mt19937 rng(random_device{}());
5 int rd(int l, int r) {
    return uniform_int_distribution<int>(l, r)(rng);
5 }
```

5 **Wzorki (2)**

2.1 Fibonacci

$f_0 = 0, f_1 = 1, f_n = f_{n-1} + f_{n-2}$, dla $n \geq 2$.

$$f_n = \frac{\alpha^n - \beta^n}{\sqrt{5}} = \frac{\alpha^n - \beta^n}{\alpha - \beta},$$

dla $\alpha = \frac{1+\sqrt{5}}{2}, \beta = \frac{1-\sqrt{5}}{2}$. Jako, że $|\beta| < 1$ to

$$f_n \approx \frac{1}{\sqrt{5}}\alpha^n.$$

Niech $F(x) = \sum_{n=0}^{\infty} f_n x^n$, wtedy

$$F(x) = \frac{x}{1 - x - x^2}.$$

$$\text{NWD}(f_n, f_m) = f_{\text{NWD}(n, m)}.$$

2.2 Liczby Catalana

$$C_n = \frac{1}{n+1} \binom{2n}{n}$$

$$C_n = C_0 C_{n-1} + C_1 C_{n-2} + \dots + C_{n-2} C_1 + C_{n-1} C_0$$

$$C_{n+1} = \frac{2(2n+1)}{n+2} C_n$$

2.3 Funkcje tworzące

$$\frac{1}{1-ax} = \sum a^n x^n$$

$$F(x) = \frac{x}{1-x-x^2}$$

$$C(x) = \frac{1-\sqrt{1-4x}}{2x}$$

$$\exp(x) = \sum_{n \geq 0} \frac{x^n}{n!}$$

$$-\ln(1-x) = \sum_{n \geq 1} \frac{x^n}{n}$$

$$\frac{1}{(1-x)^k} = \sum_{n \geq 0} \binom{n+k-1}{n} x^n$$

$$\prod_{n \geq 1} (1-x^n) = 1 + \sum_{k \geq 1} (-1)^k \left(x^{\frac{k(3k+1)}{2}} + x^{\frac{k(3k-1)}{2}} \right)$$

Matma (3)

ext-euclid.cpp

Opis: $O(\log n)$. Zwraca trójkę $[d, x, y]$, że $d = \text{NWD}(a, b)$ i $ax + by = d$.

```
tuple<ll, ll, ll> euclid(ll a, ll b) {
    if (b == 0) return make_tuple(a, 1, 0);
    auto [d, x, y] = euclid(b, a%b);
    return make_tuple(d, y, x - (a/b) * y);
}
```

fft.cpp

Opis: Mnożenie wielomianów w $O(n \log n)$.

```
using ld = long double;
using cd = complex<long double>;
const double PI = acos(-1);
```

```
void fft(vector<cd> &a, int n, bool invert = false) {
    assert((n&(n-1)) == 0);
    a.resize(n);
    for (int i = 1, j = 0; i < n; ++i) {
        int bit = n/2;
        for ( ; bit&j; bit /= 2)
            j ^= bit;
        j ^= bit;

        if (i < j) swap(a[i], a[j]);
    }
```

```
for (int len = 2; len <= n; len *= 2) {
    ld ang = 2 * PI / len * (invert ? -1 : 1);
    cd r_len(cos(ang), sin(ang));
    for (int i = 0; i < n; i += len) {
        cd r(1);
        for (int j = 0; j < len/2; ++j, r *= r_len) {
            cd u = a[i + j], v = r * a[i + j + len/2];
            a[i+j] = u + v;
            a[i+j+len/2] = u - v;
        }
    }
}
```

```
if (invert)
    for (cd &x : a) x /= n;
}
```

```
vector<ld> conv(vector<ld> a, vector<ld> b) {
    if (a.empty() || b.empty()) return {};

    vector<cd> fa(all(a)), fb(all(b));
    int n = ssize(a) + ssize(b) - 1;
```

```
int sz = 1;
while (sz < ssize(a) + ssize(b) - 1) sz *= 2;
fft(fa, sz);
fft(fb, sz);
for (int i = 0; i < sz; ++i) fa[i] *= fb[i];
fft(fa, sz, true);

vector<ld> result(n);
for (int i = 0; i < n; ++i)
    result[i] = round(fa[i].real());
return result;
}
```

mod.cpp
Opis: operacje mod 998244353.

```
const int mod = 998244353;

int add(int a, int b) {
    a += b;
    return a >= mod ? a-mod : a;
}

int sub(int a, int b) {
    return add(a, mod-b);
}

int mul(int a, int b) {
    return int((ll)a*b)%mod;
}

int fpow(int a, int b) {
    int res = 1;
    while (b) {
        if (b&1) {
            res = mul(a, res);
            b--;
        }
        a = mul(a, a);
        b /= 2;
    }
    return res;
}

int inv(int x) {
    return fpow(x, mod-2);
}
```

ntt.cpp
Opis: O(n log n). Mnożenie wielomianów mod 998244353.

```
#include "mod.cpp" //keep-include

const int g = 3;
const int g_1 = 332748118;

using vi = vector<int>;

void ntt(vi &a, int n, bool invert = false) {
    assert((n&(n-1)) == 0);
    a.resize(n);
    for (int i = 1, j = 0; i < n; ++i) {
        int bit = n/2;
        for ( ; bit&j; bit /= 2)
            j ^= bit;
        j ^= bit;

        if (i < j) swap(a[i], a[j]);
    }
```

```
for (int len = 2; len <= n; len *= 2) {
    int r_len = fpow(invert ? g_1 : g, (mod-1) / len);
    for (int i = 0; i < n; i += len) {
        int r = 1;
        for (int j = 0; j < len/2; ++j, r = mul(r, r_len)) {
            int u = a[i + j], v = mul(r, a[i + j + len/2]);
            a[i+j] = add(u, v);
            a[i+j+len/2] = sub(u, v);
        }
    }
}

if (invert) {
    int inv_n = inv(n);
    for (int &x : a) x = mul(x, inv_n);
}

vi conv(vi a, vi b) {
    if (a.empty() || b.empty()) return {};
    int n = ssize(a) + ssize(b) - 1;
    int sz = 1;
    while (sz < ssize(a) + ssize(b) - 1) sz *= 2;
    ntt(a, sz);
    ntt(b, sz);
    for (int i = 0; i < sz; ++i) a[i] = mul(a[i], b[i]);
    ntt(a, sz, true);
    a.resize(n);
    return a;
}
```

poly.cpp
Opis: Operacje na wielmianach mod 998244353. Można przepisać co się chce, przy funkcjach jest napisane co potrzeba. deriv, integr w O(n). inv, log, exp w O(n log n).

```
#include "ntt.cpp" //keep-include

using vi = vector<int>;
vi mod_xn(const vi &v, int n) { //ZAWSZE
    return vi(v.begin(), v.begin() + min(n, ssize(v)));
}

vi add(vi a, const vi &b) {
    a.resize(max(ssize(a), ssize(b)));
    for (int i = 0; i < ssize(b); ++i)
        a[i] = add(a[i], b[i]);
    return a;
}

vi sub(vi a, const vi &b) { //ZAWSZE
    a.resize(max(ssize(a), ssize(b)));
    for (int i = 0; i < ssize(b); ++i)
        a[i] = sub(a[i], b[i]);
    return a;
}

vi deriv(vi p) {
    for (int i = 1; i < ssize(p); ++i) p[i] = mul(p[i], i);
    if (ssize(p)) p.erase(p.begin());
    return p;
}

vi integr(vi p) {
    for (int i = 0; i < ssize(p); ++i)
        p[i] = mul(p[i], inv(i+1));
    p.insert(p.begin(), 0);
    return p;
}
```

```
}

vi inv(vi p, int n) {
    assert(ssize(p) && p[0] != 0);
    if (n == 1)
        return {inv(p[0])};
    p.resize(n);
    vi _p = p;
    for (int i = 1; i < n; i += 2) _p[i] = sub(0, p[i]);

    vi t = mod_xn(conv(p, _p), n);
    for (int i = 1; i < n; i += 2) assert(t[i] == 0);
    for (int i = 0; i < n; i += 2)
        t[i/2] = t[i];
    t = inv(t, (n+1)/2);
    t.resize(n);
    for (int i = (n-1)/2; i >= 0; --i) {
        t[2*i] = t[i];
        if(i) t[i] = 0;
    }
    return mod_xn(conv(_p, t), n);
}

vi log(const vi &p, int n) { //DERIV, INV, INTEGR
    assert(ssize(p) && p[0] == 1);
    return integr(mod_xn(conv(deriv(mod_xn(p, n)), inv(p, n)), n
        -1));
}

vi exp(const vi &p, int n) { //ADD, LOG
    assert(p.empty() || p[0] == 0);
    vi q = {1};
    for (int x = 1; x < n; x *= 2)
        q = mod_xn(conv(q, add(sub({1}, log(q, 2*x)), mod_xn(p, 2*x
            )), 2*x);
    q.resize(n);
    return q;
}

vi powi(vi p, int k, int n) { //LOG, EXP
    p = mod_xn(p, n);
    int cnt = 0;
    while(cnt < ssize(p) and !p[cnt])
        ++cnt;
    if(ll(cnt) * k >= n)
        return {};
    p.erase(p.begin(), p.begin() + cnt);
    if(p.empty())
        return k ? vi{} : vi{1};
    int powi0 = fpow(p[0], k);
    int inv0 = inv(p[0]);
    for (int &x : p) x = mul(x, inv0);
    p = log(p, n - cnt * k);
    for (int &x : p) x = mul(x, k);
    p = exp(p, n - cnt * k);
    for (int &x : p) x = mul(x, powi0);
    vi t(cnt * k, 0);
    p.insert(p.begin(), t.begin(), t.end());
    return p;
}

vi sqrt(const vi &p, int n) { //POW
    assert(ssize(p) && p[0] == 1);
    return powi(p, inv(2), n);
}
```

poly2d.cpp

Opis: Operacje na wielmianach dwóch zmiennych mod 998244353. Mnożenie w $O(mn \log(m+n))$. Dzielenie zwraca vector współczynników przy x^{n-1} , w czasie $O(mn \log^2(n+m))$.

```
#include "poly.cpp" //keep-include

void ntt(vector<vi> &a, int n, int m, bool inverse = false) {
    a.resize(n);
    for (int i = 0; i < ssize(a); ++i)
        ntt(a[i], m, inverse);
    for (int j = 0; j < ssize(a[0]); ++j) {
        vi c;
        for (int i = 0; i < ssize(a); ++i)
            c.eb(a[i][j]);
        ntt(c, n, inverse);
        for (int i = 0; i < ssize(a); ++i)
            a[i][j] = c[i];
    }
}

vector<vi> conv(vector<vi> a, vector<vi> b, int n, int m) {
    if (a.empty() || b.empty()) return {};
    int l1 = ssize(a) + ssize(b) - 1, sz1 = 1 << __lg(2*l1-1);
    int l2 = ssize(a[0]) + ssize(b[0]) - 1, sz2 = 1 << __lg(2*l2-1);

    ntt(a, sz1, sz2);
    ntt(b, sz1, sz2);
    for (int i = 0; i < sz1; ++i)
        for (int j = 0; j < sz2; ++j) a[i][j] = mul(a[i][j], b[i][j]);
    ntt(a, sz1, sz2, true);
    a.resize(n);
    for (int i = 0; i < n; ++i) a[i].resize(m);
    return a;
}

vi div(vector<vi> b, vector<vi> a, int n, int m) {
    if (n == 1) {
        return conv(b[0], inv(a[0], m));
    }
    vector<vi> q = a;
    for (int i = 1; i < n; i += 2)
        for (int j = 0; j < ssize(a[i]); ++j) q[i][j] = sub(0, q[i-1][j]);

    vector<vi> v = conv(a, q, n, 2*m);
    debug(v);
    q = conv(b, q, n, 2*m);
    for (int i = 0; i < ssize(v); i += 2) {
        for (int j = 0; j < ssize(v[i]); ++j) {
            v[i/2][j] = v[i][j];
        }
    }
    v.resize((n+1)/2);

    if (n%2 == 0) {
        for (int i = 1; i < n; i += 2)
            for (int j = 0; j < 2*m; ++j)
                q[i/2][j] = q[i][j];
        q.resize(n/2);
        return div(q, v, n/2, 2*m);
    }
    else {
        for (int i = 0; i < n; i += 2)
            for (int j = 0; j < 2*m; ++j)
                q[i/2][j] = q[i][j];
        q.resize((n+1)/2);
```

```
        return div(q, v, (n+1)/2, 2*m);
    }
}
```

Struktury danych (4)

wavelet.cpp

Opis: $O(\log^2 \text{ALPH})$ czasowo, $O(n \log \text{ALPH})$ pamięciowo. Constructor przyjmuje ciąg na którym rozpinamy drzewo, rank(l, r, k) zwraca k-tą najmniejszą liczbę na przedziale [l; r]. (**Uwaga!**) pamięć około 262 Mb, dla $N, Q \leq 2 \cdot 10^5$.

```
//funkcja cnt jest w O(\log) co daje cale operacje w O(\log^2),
//da sie cnt w czasie stalym — do dopisania
struct wavelet_tree {
    const static int ALPH = (1<<18)-1;
    struct Node {
        int low, high, mid;
        vector<int> v;
        Node *left, *right;

        Node() {
            low = 0, high = ALPH;
            left = right = nullptr;
            mid = (low+high)/2;
        }

        Node (const vector<int> &x, const vector<int> &t, int a = 0, int b = ALPH): low(a), high(b), v(x) {
            mid = ((l1)low+high)/2;
            if(ssize(x) && low != high) {
                vector<int> div[2];
                for(auto e : v) div[t[e]<=mid].eb(e);
                if(ssize(div[1])) left = new Node(div[1], t, low, mid);
                else left = nullptr;
                if(ssize(div[0])) right = new Node(div[0], t, mid+1, high);
                else right = nullptr;
            }
            else {
                left = right = nullptr;
            }
        }
    };

    Node *root;
    wavelet_tree(const vector<int> &b) {
        vector<int> idx;
        REP(i, ssize(b)) idx.eb(i);
        root = new Node(idx, b);
    };

    int cnt(Node *v, int l, int r) {
        if(v == nullptr) return 0;
        return int(upper_bound(all(v->v), r)-lower_bound(all(v->v), l));
    }

    int rank(int l, int r, int k, Node *v) {
        if(v->low == v->high) return v->low;
        if(cnt(v->left, l, r) >= k) return rank(l, r, k, v->left);
        else return rank(l, r, k-cnt(v->left, l, r), v->right);
    }

    int rank(int l, int r, int k) {
        return rank(l, r, k, root);
    }

    void clear(Node *v) {
```

```
        if(v->left != nullptr) clear(v->left);
        if(v->right != nullptr) clear(v->right);
        delete v;
    }
    ~wavelet_tree() {
        clear(root);
    }
};
```

treap.cpp

Opis: $O(\log n)$ czasowo. Treap z operacjami na (multi)secie. kth(t, k) zwraca k-tą najmniejszą liczbę.

```
namespace Treap {
    mt19937 rng(420);
    struct Node {
        Node *left, *right;
        int val, rank;
        int size;
        Node(): left(nullptr), right(nullptr), val(0), rank(int(rng
            ())), size(0) {}

    };
    using pNode = Node*;

    Node pool[1'000'003];
    int curr = 0;

    int sz(pNode t) {
        return t == nullptr ? 0 : t->size;
    }

    void upd(pNode t) {
        t->size = sz(t->left) + sz(t->right) + 1;
    }

    pNode merge(pNode a, pNode b) {
        if(a == nullptr) return b;
        if(b == nullptr) return a;

        if(a->rank < b->rank) {
            a->right = merge(a->right, b);
            upd(a);
            return a;
        }
        else {
            b->left = merge(a, b->left);
            upd(b);
            return b;
        }
    }

    pair<pNode, pNode> split(pNode t, int x) {
        if(t == nullptr) return {nullptr, nullptr};

        if(t->val <= x) {
            auto [a, b] = split(t->right, x);
            t->right = a;
            upd(t);
            return {t, b};
        }
        else {
            auto [a, b] = split(t->left, x);
            t->left = b;
            upd(t);
            return {a, t};
        }
    }

    void insert(pNode &root, int x) {
```

```
pNode u = pool + (curr++);
u->val = x;
u->size = 1;
auto [a, b] = split(root, x);
root = merge(merge(a, u), b);
}

bool find(pNode t, int x) {
    if(t == nullptr) return false;
    if(x == t->val) return true;
    if(x < t->val) return find(t->left, x);
    else return find(t->right, x);
}

void erase(pNode root, int x) {
    auto [ab, c] = split(root, x);
    auto [a, b] = split(ab, x-1);
    root = merge(a, c);
}

int kth(pNode t, int k) {
    assert(k >= 0);
    if(sz(t->left) == k-1) return t->val;
    if(k > sz(t->left)) return kth(t->right, k-1-sz(t->left));
    else return kth(t->left, k);
}
}
```

implicit-treap.cpp

Opis: $O(\log n)$ czasowo. Implicit treap z podstawowymi operacjami, przykładowo odwracanie na przedziale (może być dowolna operacja). insert(root, k, x) wstawia element o wartości x w k -te miejsce tablicy. rev(root, l, r) odwraca przedział $[l; r]$ (indeksy od 0).

```
namespace Treap {
    mt19937 rng(420);
    struct Node {
        Node *left, *right;
        int rank;
        int size;
        bool rev;
        int val;
        Node(): left(nullptr), right(nullptr), rank(int(rng())),
            size(0), rev(false), val(0) {}
    };
    using pNode = Node*;

    Node pool[300'003];
    int curr = 0;

    int sz(pNode t) {
        return t == nullptr ? 0 : t->size;
    }

    void upd(pNode t) {
        t->size = sz(t->left) + sz(t->right) + 1;
    }

    void push_lazy(pNode t) {
        if(t->rev == false) return;
        if(t->left) t->left->rev = !t->left->rev;
        if(t->right) t->right->rev = !t->right->rev;
        swap(t->left, t->right);
        t->rev = false;
    }

    pNode merge(pNode a, pNode b) {
        if(a == nullptr) return b;
        if(b == nullptr) return a;
```

```
push_lazy(a);
push_lazy(b);
if(a->rank < b->rank) {
    a->right = merge(a->right, b);
    upd(a);
    return a;
}
else {
    b->left = merge(a, b->left);
    upd(b);
    return b;
}
}

pair<pNode, pNode> split(pNode t, int x) {
    if(t == nullptr) return {nullptr, nullptr};

    push_lazy(t);
    if(sz(t->left)+1 <= x) {
        auto [a, b] = split(t->right, x-(sz(t->left)+1));
        t->right = a;
        upd(t);
        return {t, b};
    }
    else {
        auto [a, b] = split(t->left, x);
        t->left = b;
        upd(t);
        return {a, t};
    }
}

pNode build(vector<int> a) {
    pNode root = nullptr;
    int n = ssize(a);
    for(int i = 0; i < n; ++i) {
        pNode u = pool + i;
        u->size = 1;
        u->val = a[i];
        root = merge(root, u);
    }
    curr += n;
    return root;
}

void insert(pNode &root, int k, int x) {
    auto [a, b] = split(root, k-1);
    Node *u = pool + (curr++);
    u->size = 1;
    u->val = x;
    root = merge(merge(a, u), b);
}

void reverse(pNode &root, int l, int r) {
    auto [ab, c] = split(root, r+1);
    auto [a, b] = split(ab, l);
    b->rev = !b->rev;
    root = merge(a, merge(b, c));
}

void print(pNode t) {
    if(t == nullptr) return;
    push_lazy(t);
    print(t->left);
    cout << t->val << ' ';
    print(t->right);
}
}
```

uf-rollback.cpp

Opis: $O(\log n)$ czasowo. Union find z roll backami – roll_back(x) cofa ostatnie x operacji unionn (union wierzchołków z tego samego zbioru się nie liczy).

```
struct dynamicUF {
    vector<int> e;
    dynamicUF(int n):e(n, -1) {}
    stack<array<int, 3>> s;

    int get(int a) {
        while(e[a] >= 0) a = e[a];
        return a;
    }

    int unionn(int a, int b) {
        a = get(a); b = get(b);
        if(a == b) return 0;
        if(-e[a] > -e[b]) swap(a, b);
        s.push({a, b, e[a]});
        e[b] += e[a];
        e[a] = b;
        return 1;
    }

    void roll_back(int m = 1) {
        while (m--) {
            auto [a, b, sz] = s.top();
            s.pop();
            e[b] -= sz;
            e[a] = sz;
        }
    }
};
```

fenwick-tree.cpp

Opis: (Binary Indexed Tree) $O(\log N)$, indeksowane od 0. Update w punkcie, zapytanie na przedziale

```
struct BIT {
    vector<int> tree;
    int N;
    BIT() {N = 0};
    BIT(int n) {
        N = n;
        tree.resize(N);
    }
    void resize(int n) {
        N = n;
        tree.resize(N);
    }
    void update(int p, int x) {
        for (++p; p <= N; p += (p&-p)) tree[p-1] += x;
    }
    int sum(int r) {
        int res = 0;
        for (; r > 0; r -= (r&-r)) res += tree[r-1];
        return res;
    }
    int query(int l, int r) {return sum(r+1)-sum(l);}
    int lower_bound(int sum) {
        int pos = 0;
        for (int pw = 1<<25; pw; pw>=1) {
            int npos = pos+pw;
            if (npos <= N && tree[npos-1] < sum) {
                pos = npos; sum -= tree[pos-1];
            }
        }
        return pos;
    }
};
```

```
};
```

Grafy (5)

hld.cpp

Opis: $O(n)$ preprocessing, update i query w $O(\log n)$. Update to zmiana wartości w wierzchołku, query to max na ścieżce. Drzewo przedziałowe do dopisania tak jak w komentarzu.

```
const int inf = 1e9+10;

//struct segtree {
//    segtree(int n = 0) {}
//    void update(int v, int x) {}
//    int query(int l, int r) {}
//};

struct HLD {
    int n;
    vector<vector<int>>> g;
    segtree t;
    int timer;

    vector<int> sz, d, parent, path, c, pre;

    HLD(const vector<vector<int>>> &graph): n(ssize(graph)), g(
        graph) {
        sz = d = parent = path = c = pre = vector<int> (n);

        d[0] = 0;
        dfs(0);

        path[0] = 0;
        timer = 0;
        make_hld(0);
        t = segtree(timer);
    }

    void dfs(int v, int p = -1) {
        parent[v] = p;
        sz[v] = 1;
        c[v] = -1;
        for (auto u : g[v]) {
            if (u == p) continue;
            d[u] = d[v] + 1;
            dfs(u, v);
            sz[v] += sz[u];
            if (c[v] == -1 || sz[c[v]] < sz[u]) c[v] = u;
        }
    }

    void make_hld(int v, int p = -1) {
        pre[v] = timer++;
        if (c[v] != -1) {
            path[c[v]] = path[v];
            make_hld(c[v], v);
        }
        for (auto u : g[v]) {
            if (u == p || u == c[v]) continue;
            path[u] = u;
            make_hld(u, v);
        }
        //post[v] = timer++;
    }

    void update(int v, int x) {
        t.update(pre[v], x);
    }
}
```

```
int query(int a, int b) {
    int res = -inf;
    while (path[a] != path[b]) {
        if (d[path[a]] < d[path[b]]) swap(a, b);
        res = max(res, t.query(pre[path[a]], pre[a]));
        a = parent[path[a]];
    }
    if (d[a] < d[b]) swap(a, b);
    res = max(res, t.query(pre[b], pre[a]));
    return res;
}
};
```

Tekstówki (6)

z.cpp

Opis: $O(n)$. Zwraca tablice z, gdzie $z[i]$ jest równe najdłuższemu wspól-nemu prefiksowi słowa s i sufixu $[i, \dots, n - 1]$.

```
vector<int> z(string s) {
    int n = ssize(s);
    vector<int> z(n);

    int l = 0, r = 0;
    for (int i = 1; i < n; ++i) {
        if (i < r) z[i] = min(z[i-1], r-i);
        while (i + z[i] < n && s[i+z[i]] == s[z[i]]) ++z[i];
        if (i + z[i] > r) {
            r = i + z[i];
            l = i;
        }
    }
    return z;
}
```

suffix-array.cpp

Opis: (źródło: <https://github.com/tonowak/acmlib/blob/master/code/string/array-short/main.cpp>) $O(n \log n)$, zawiera posortowane suffixy, zawiera pusty suffix, $lcp[i]$ to lcp suffixu $sa[i - 1]$ i $sa[i]$. Dla $s = aabaaab$, $sa = \{7, 3, 4, 0, 5, 1, 6, 2\}$, $lcp = \{0, 0, 2, 3, 1, 2, 0, 1\}$

```
pair<vector<int>, vector<int>>> suffix_array(vector<int> s, int
    alpha = 26) {
    ++alpha;
    for(int &c : s) ++c;
    s.emplace_back(0);
    int n = ssize(s), k = 0, a, b;
    vector<int> x(s.begin(), s.end());
    vector<int> y(n), ws(max(n, alpha)), rank(n);
    vector<int> sa = y, lcp = y;
    iota(sa.begin(), sa.end(), 0);
    for(int j = 0, p = 0; p < n; j = max(1, j * 2), alpha = p) {
        p = j;
        iota(y.begin(), y.end(), n - j);
        for (int i = 0; i < n; ++i) if(sa[i] >= j)
            y[p++] = sa[i] - j;
        fill(ws.begin(), ws.end(), 0);
        for (int i = 0; i < n; ++i) ws[x[i]]++;
        for (int i = 1; i <= alpha - 1; ++i) ws[i] += ws[i - 1];
        for(int i = n; i--;) sa[--ws[x[y[i]]]] = y[i];
        swap(x, y);
        p = 1, x[sa[0]] = 0;
        for (int i = 1; i <= n-1; ++i) a = sa[i - 1], b = sa[i], x[
            b] =
                (y[a] == y[b] && y[a + j] == y[b + j]) ? p - 1 : p++;
    }
    for(int i = 1; i <= n-1; ++i) rank[sa[i]] = i;
```

```
for(int i = 0, j; i < n - 1; lcp[rank[i++]] = k)
    for(k && k--, j = sa[rank[i] - 1];
        s[i + k] == s[j + k]; k++);
lcp.erase(lcp.begin());
return {sa, lcp};
}
```

Optymalizacje (7)

fastio.cpp

Opis: fastio.

```
bool is_digit(char c) {
    return c >= '0';
}

int fastin(){
    int x = 0;
    char c;
    while (is_digit(c = getchar_unlocked())) {
        x = x * 10 + c - '0';
    }
    return x;
}

void write_int(unsigned long x) {
    char t[19];
    int i = 0;
    do {
        int d = x%10;
        t[i++] = '0'+d;
        x /= 10;
    } while (x > 0) ;

    while (--i >= 0) {
        putchar_unlocked(t[i]);
    }
    putchar_unlocked('\n');
```